

Investments in Information Systems following Global Corporate Sustainability Due Diligence Directives

**Report for the CentER Honors Research Experience
May 12th, 2026**

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Abstract

Globalization has fundamentally reshaped how goods are produced and consumed, creating vast networks that span the entire world. While this shift has increased efficiency, it has also led to a significant "accountability gap" where firms lack visibility into working conditions and environmental impacts deep within their multi-tiered supply chains. Research indicates that earlier voluntary Corporate Social Responsibility efforts largely failed to produce meaningful change, as they often provided only a narrow image of a workplace and allowed companies to ignore deeper risks. Consequently, policy makers have shifted from voluntary guidelines toward stricter legal mandates.

Early models, such as the California Transparency in Supply Chains Act (CTSCA) and the UK's Modern Slavery Act, function on "passive disclosure". These laws rely on public shaming, consumer choice, and investor pressure rather than direct legal penalties. In contrast, modern European frameworks, such as the French Duty of Vigilance law, the German Supply Chain Act, and the EU's Corporate Sustainability Due Diligence Directive (CSDDD), represent an "active mitigation" model. These directives legally compel businesses to actively identify, prevent, and fix abuses across their entire network, including indirect suppliers. Failure to comply now results in severe consequences, including massive financial penalties tied to global revenue and civil liability in European courts.

To meet these rigorous standards, companies must transform their governance by investing in advanced information systems. Information technology is a primary driver of supply chain governance, allowing firms to monitor partner behavior and coordinate complex activities. We utilize the Computer Intelligence Database from Aberdeen, which provides high-quality data on specific hardware and software deployments validated through a commercial feedback loop. By analyzing a hierarchy of 224 model groups across 10 core business activities, we test how these tools can be organized to meet compliance needs. Our work identifies seven specific software categories essential for companies affected by these mandates: Team/Committee Communication for internal task force coordination, Audit software to act as an official system of record for facility inspections, Evaluation and Compliance to manage mandatory certifications, Connecting Businesses in the Supply Chain to provide deep-tier visibility, Information Sharing for the secure exchange of corporate policies, HR and CRM to track employee training and manage external public inquiries, Generating Reports to synthesize data into legally required disclosure formats. Ultimately, the report concludes that the Computer Intelligence Database is a feasible source for answering the core research questions. We emphasize that successful compliance requires deep digital synchronization and the integration of manufacturing processes, supply networks, and internal communications to ensure transparency and audit readiness.

1 Introduction and Problem Definition

Globalization has fundamentally reshaped how goods are produced, transported and consumed, creating vast networks that span the entire world. While this shift has allowed companies to lower costs and increase efficiency, it has also led to a significant "accountability gap" (Villiers, 2022). In many cases, the complexity of these multi-tiered supply chains has left lead firms with a lack of visibility regarding the working conditions and environmental impacts occurring deep within their operations. As production moved to countries with weaker labor protections, companies often lost track of how their business activities affected local communities and ecosystems (De Schutter, 2020).

Our broad goal is to answer the following two research questions:

1. Do firms implement supply chain reporting/risk management information systems after the introduction of a supply chain reporting mandate?
2. Do firms integrate their information systems across supply chains after the introduction of a supply chain reporting mandate?

At first, the primary response to these issues was implementing voluntary Corporate Social Responsibility (CSR) initiatives. Companies were encouraged to develop their own codes of conduct and monitor their internal activity. However, research indicates that these voluntary efforts have largely failed to produce meaningful change. Smit et al. (2021) argue that common voluntary tools, such as traditional social auditing, are often superficial, providing only a narrow image of a workplace and failing to capture human rights violations, such as forced labor. Furthermore, because disclosure was voluntary, many companies only reported positive information while ignoring the deeper risks in their supply chains (Villiers, 2022). Because of these issues, policy makers have shifted the focus from voluntary guidelines to stricter law mandates.

Even within these law mandates, we see a shift from simple transparency requirements to strict legal obligations. For example, early frameworks, known as disclosure-based directives, only ask companies to publicly report on their practices. The list includes the UK Modern Slavery Act, which relies on public pressure rather than direct financial penalties (UK Parliament, 2015). In contrast, modern laws, such as the French Corporate Duty of Vigilance and the upcoming EU Corporate Sustainability Due Diligence Directive, compel businesses to actively identify, prevent, and fix human rights and environmental issues, with failures resulting in massive fines or lawsuits (Gérard-Rodriguez, 2025; European Commission, n. d.). Furthermore, governments are increasingly enforcing market access regulations, such as the US Uyghur Forced Labor Prevention Act and the EU Deforestation Regulation, to physically block or seize questionable goods at the border so they never reach consumers (U.S. Customs and Border Protection, 2025; European Commission, n.d.). Finally, sector-specific directives, like the EU Conflict Minerals Regulation, apply these rigorous checks directly to high-risk materials, creating an overall more accountable global market (European Commission, n.d.).

To understand how companies respond to modern regulatory pressures, it is helpful to look at the real effects of accounting disclosures. According to Kanodia and Sapra (2016), public reporting is not a simple administrative exercise, but a powerful tool that changes how managers behave. Because administrators anticipate that investors, consumers, and NGOs will judge their public reports, they often alter their operations long before the report is drafted. In the context of supply chain due diligence, this means management may terminate non-compliant suppliers or redesign entire logistics networks specifically to ensure their future public disclosures meet both legal requirements and societal expectations. However, the decision to be transparent is complicated by the proprietary cost of disclosure. While revealing supply chain details can prove a firm's commitment to ethics and lower its cost of capital, it also creates a risk of revealing sensitive trade secrets to competitors (Kanodia, 2007). Providing granular information about supplier identities or production capacities could allow rivals to exploit a firm's unique logistics advantages, leading to a loss of competitive advantage.

Moreover, to meet these rigorous legal standards, companies must transform how they govern their supply chains by investing in information systems. Dolci and Maçada (2014) demonstrate that information technology is a primary driver of supply chain governance, as it allows firms to better coordinate activities and monitor partner behavior. Furthermore, the limitations of traditional audits have created a demand for more

advanced technological solutions. Leading companies are now exploring advanced governance tools, such as worker-voice applications and blockchain-enabled tracking, to gather real-time data directly from the source (Smit et al., 2021). These systems allow for a continuous flow of information that can alert a company to risks more efficiently. As sustainability becomes a core requirement for remaining competitive in the global market, the ability to manage complex data through information systems has become essential (Villiers, 2022).

The objectives of this report are:

1. Understanding the current setting on supply chain due diligence by reviewing the existing academic literature
2. Describing the current regulatory frameworks, focusing on the California Transparency in Supply Chains Act
3. Exploring the Computer Intelligence Database and assessing whether or not it is feasible to answer the core research questions using this data source

The importance of these objectives lies in their ability to bridge the gap between theoretical knowledge and practical application within the evolving field of supply chain due diligence. By first conducting a comprehensive review of academic literature, the project establishes a firm foundation. This step is essential because it allows the study to build upon existing insights while identifying specific gaps that still need to be addressed.

Furthermore, focusing on regulatory frameworks provides the necessary legal context that drives corporate accountability. As global markets shift toward greater transparency, understanding how specific laws influence business behavior is vital. By analyzing these regulations, the study moves beyond abstract ideas and looks at the actual rules that govern how products are made and moved across the globe.

Finally, the exploration of the Computer Intelligence Database serves as the critical bridge to empirical evidence. Assessing the feasibility of this database at the outset is a necessary safeguard, as it ensures that the project's conclusions are based on high-quality, actionable information. Together, these three objectives form a clear, step-by-step path: they provide the context, define the legal boundaries, and verify the tools required to produce a rigorous study. By addressing these areas, the project ensures its results are both academically sound and practically useful for anyone navigating the complexities of modern supply chain management.

2 The California Transparency in Supply Chains Act

We focus on the California Transparency in Supply Chains Act (CTSCA), which took effect on January 1st, 2012. The law requires large retailers and manufacturers, specifically those with over 100 million dollars in global revenue, to publicly disclose what they are doing to eliminate slavery and human trafficking from their supply chains (California Department of Justice, Office of the Attorney General, 2012).

Because California is a state, it doesn't have the legal authority under international trade laws to ban goods made with forced labor. Instead, the legislation relies on public pressure and reputation risk. By forcing companies to be transparent, the law gives consumers and investors the information they need to choose ethical brands over brands that violate human rights. Because of this, a company could legally comply simply by posting a clear statement on its website admitting that it does absolutely nothing to stop forced labor. However, since disclosing such a statement would have reputational consequences, the law effectively incentivizes major companies to take real actions regarding their supply chain risks. The legislation requires applicable organizations to disclose the extent of their activities across five highly specific operational domains, which are explored in detail in Table 1 (California Department of Justice, Office of the Attorney General, 2012).

When the new mandate took effect, companies had to completely rethink how they viewed their supply chains. In the past, corporate software was built to track simple, linear relationships, mainly focusing on Tier 1 suppliers where the actual monetary transactions were taking place. However, under the new

legislation, companies had to start mapping their networks much more deeply. Corporate leaders realized that if they lacked visibility into their Tier 2 and Tier 3 suppliers, they couldn't accurately report on human rights risks (Carter et al., 2015).

To handle this reality, major corporations broke down their internal silos. Instead of leaving supply chain ethics to just one department, they built integrated compliance teams that pulled together professionals from HR, procurement, legal, and corporate social responsibility (Pagell and Wu, 2009). This required a major technology upgrade, too. Companies had to replace disconnected departmental databases with shared platforms, ensuring that if a supply chain manager flags a labor issue on the ground, the company's legal team can see that data instantly (Foerstl et al., 2015).

Disclosure Area	Legal Requirement	Technological Needs
Verification	Companies must disclose how they check their supply chains for human trafficking risks, and whether they use a third party to do it.	Risk-mapping software and data tools to flag high-risk regions and materials.
Audits	Companies must report if they audit suppliers for anti-slavery compliance, and whether those audits are independent and unannounced.	Audit-tracking databases, mobile apps for field inspectors, and systems to track when suppliers need to fix issues.
Certification	Direct suppliers must certify that their materials follow the anti-slavery laws of the countries where they operate.	Automated vendor compliance portals, digital contract management, and e-signature databases.
Internal Accountability	Companies must explain their internal anti-trafficking rules and detail how they handle employees or contractors who break them.	Softwares for linking supply chain compliance data with HR systems and internal whistleblower or incident-reporting platforms.
Training Programs	Companies must share how they train supply chain managers to identify and stop human trafficking.	Learning Management Systems to track and verify that employees have completed their training.

Table 1: Disclosure Domains and Database Implications

3 European Counterparts

We will now further discuss how such acts are phrased and implemented in Europe, as a direct comparison to the United States. The European Union is currently pushing a more advanced regulatory framework with the Corporate Sustainability Due Diligence Directive (CSDDD). Unlike earlier transparency laws that required companies to report on their supply chains, the CSDDD is an operational mandate. It specifically targets enterprises having over 1,000 employees and a global turnover exceeding €450 million, with a phased roll-out between 2028 and 2029. Under this directive, it is not enough for companies to simply publish a compliance statement. They are legally required to actively track down, prevent, and fix environmental and human rights abuses across their entire network, including indirect suppliers. The enforcement mechanisms are severe: non-compliant companies face fines of up to 5 percent of their worldwide turnover. Most importantly, the law introduces civil liability, meaning victims of corporate abuse can directly sue these multinational companies for damages in European courts (European Commission, n. d.).

Running parallel to these efforts is the Corporate Sustainability Reporting Directive (CSRD). While the CSDDD focuses on operational changes, the CSRD is a rigorous disclosure law. It casts a wider net, applying to any large EU company that meets two of the following three criteria: over 250 employees, €50 million in net turnover, or €25 million in total assets. It also captures non-EU companies that do significant business within the European block. To prevent vague or misleading disclosures, all data must be formatted using the specific European Sustainability Reporting Standards. Enforcement is handled by individual member states, typically through financial fines (European Commission, n. d.).

Before the broader European Union directives were introduced, France led the way with its pioneering 2017 Duty of Vigilance Law (Loi de Vigilance). This legislation specifically targets major French corporations: with at least 5,000 employees inside France or 10,000 employees worldwide, including their subsidiaries. The core requirement of the law is that these companies must create, publish, and actively follow an annual "Vigilance Plan." Businesses have to clearly map out exactly how they will prevent severe environmental damage, health hazards, and human rights abuses across their own operations, their subsidiaries, and their established network of suppliers. The enforcement of this law relies heavily on public accountability. Labor unions and NGOs act as watchdogs and can formally put a non-compliant company on notice. If the business doesn't fix or implement its Vigilance Plan within three months, a judge can step in, issue legal injunctions, and impose ongoing, daily fines until the company complies. Furthermore, if a company's failure to act leads directly to preventable damage, the courts can ascribe them penalties of up to €30 million (Gérard-Rodriguez, 2025).

Another major piece of national legislation is Germany's Supply Chain Due Diligence Act (Lieferkettensorgfaltspflichtengesetz or LkSG), which officially took effect in 2023. This law targets any company that has its central administration or registered office in Germany and employs at least 1,000 people. The LkSG requires companies to establish dedicated risk management systems to catch and mitigate human rights and environmental violations. Unlike the French law, which extends deeply into the sub-tier network, the German mandate primarily focuses on a company's own operations and its Tier 1 suppliers. To ensure these rules are actually followed, companies must appoint a specific human rights officer, create a clear complaints procedure for workers, and publish an annual compliance report. Enforcement is handled by Germany's control authority, and the penalties are severe. Large companies caught violating the act can be fined up to 2 percent of their average annual global turnover. Furthermore, non-compliant businesses face a significant commercial threat: they can be completely blacklisted and excluded from lucrative public procurement contracts for up to three years (Federal Ministry of Labour and Social Affairs, 2023).

Moving to Scandinavia, Norway introduced its own Transparency Act (Åpenhetsloven) in 2022. It applies to any company selling goods or services in Norway that meets two of the following three criteria: sales revenue of 70 million NOK, a balance sheet of 35 million NOK, or an average of 50 full-time employees. Under this act, companies are required to conduct thorough due diligence to guarantee fundamental human rights and decent working conditions throughout their operations. However, the law's most groundbreaking feature is that it grants the general public a legal "right to information." This means that consumers, journalists, or NGOs can submit a written request asking a company exactly how it handles a specific human rights issue, and the business is legally obligated to respond within three weeks. Compliance is monitored by the Norwegian Consumer Authority. If a company fails to monitor its supply chain properly, or if it simply ignores inquiries from the public, regulators can step in and issue direct infringement fines and enforcement penalties (Forbrukertilsynet, 2022).

Switzerland took a much more targeted approach with its Ordinance on Due Diligence and Transparency (DDTrO), which went into effect in 2022. Rather than regulating companies based purely on their size or revenue like its European neighbors, Switzerland focuses on specific, high-risk activities. The law applies to any company based in the country that imports or processes conflict minerals (specifically tin, tantalum, tungsten, or gold) or handles products where there is a reasonable suspicion of child labor. Because the scope of the law is so narrow, the requirements are equally focused. Affected companies have to meticulously trace their supply chains and build management systems explicitly designed to catch and eliminate those two exact issues. They must also publish an annual report detailing these specific efforts. While the scope is limited, the consequences for breaking the rules are uniquely severe. If a company fails to file these reports or provides false information about its supply chain, it crosses into criminal territory, with regulators levying

criminal fines of up to 100,000 CHF (The Federal Council, 2021).

Across the Channel, the United Kingdom took an earlier, slightly different approach with its Modern Slavery Act of 2015. This legislation applies to any commercial organization doing business in the UK that generates a global annual turnover of 36 million pounds or more. Under Section 54 of the Act, these companies are required to publish an annual statement detailing the exact steps they have taken to keep slavery and human trafficking out of their operations and supply chains. While the government can seek a court injunction to force a company to publish a missing report, there are currently no financial fines for actually having poor supply chain practices. The ultimate enforcement is left entirely to the pressure of investors, consumers, and civil society (UK Parliament, 2015).

Finally, we discuss the pushback against supply chain due diligence acts, because it represents a major tension between humanitarian goals and economic concerns. For example, resistance against the European Union’s CSDDD became most visible in early 2024 when several member states blocked the original proposal. They argued that the new rules would create legal uncertainty for businesses already struggling with high energy costs. To save the directive, a series of drastic compromises were made, significantly raising the thresholds for which companies are covered. Instead of applying to mid-sized firms, the law now focuses almost exclusively on the corporate groups (DW, 2024).

This trend of “simplification” continued with the introduction of the 2025 and 2026 Omnibus legislative packages. These packages were designed to reduce the reporting burden on companies mandated by the supply chain due diligence acts. As a result, many requirements were narrowed to focus primarily on a company’s direct business partners rather than the entire supply chain, and the full implementation of the law was delayed until 2029. Opponents also raised concerns about how companies might stop doing business in developing nations to avoid legal liability, which could unintentionally harm the very workers the law was designed to protect. Consequently, the current legal landscape reflects a compromise: while the principle of mandatory due diligence remains, the scope has been narrowed to prioritize the largest global players and give more time for the market to adjust to these complex digital and legal demands (European Commission, n.d.).

Overall, when analyzing the evolution of global supply chain legislation, there is a stark contrast between early models like the California Transparency in Supply Chains Act of 2012 and the modern European frameworks we see today. The California law, much like the UK’s 2015 Modern Slavery Act, was built on the concept of passive disclosure. It simply mandates that companies publicly report what they are doing to stop forced labor, even if the honest answer is “nothing.” It relies on public shaming, consumer choices, and investor pressure to drive change, rather than imposing strict legal penalties for poor practices. In contrast, national laws in France and Germany, alongside the EU Corporate Sustainability Due Diligence Directive, no longer allow companies to simply check their work. They legally compel businesses to actively hunt down, mitigate, and prevent human rights and environmental abuses deep within their multi-tier supply chains. Furthermore, the stakes are exponentially higher. While the California mandate lacks heavy financial consequences, the European frameworks enforce compliance through massive financial penalties tied to global revenue, exclusion from public contracts, and severe civil and criminal liability.

4 Literature Review

Supply chain transparency is an important topic in operations management and corporate governance research. Studies show that while being open about supply chains can build market trust and improve sustainability, companies often choose to disclose too little information. Without strong market pressure to hold them accountable, firms default to defensive, purely symbolic compliance.

Before a company can share information with the public, it needs the right internal systems and a willingness to connect data across its own departments. External transparency can’t be achieved without internal visibility and traceability. If a firm lacks the mechanisms to track materials or human rights practices behind the scenes, genuine public transparency is impossible (Budler et al., 2024). Technologies like blockchain, and enterprise management systems definitely help track this data. However, investing in technology alone is not the end solution, as companies should also correct the identified negative practices (Budler et al., 2024).

The decision to share this data is sometimes blocked by the fear of leaking proprietary information. However, strong intellectual property laws make companies more comfortable integrating their internal systems. For example, Andreicovici, Bormann, and Hombach (2025) found that when U.S. states adopted the Uniform Trade Secrets Act, firms became much more likely to connect their internal data. Because the law lowers the financial risks of information leaking to competitors, companies feel safer sharing sensitive data across their own departments using centralized EMS platforms. This internal openness ultimately leads to real business benefits, including fewer reporting mistakes, quicker earnings announcements, and better overall decision-making (Andreicovici et al., 2025).

Once a company connects its internal data, managers face a strategic choice of how much information they should actually share with the public. They have to weigh the potential boost to their brand against the risks of exposing their supply chain. Marshall et al. (2016) map this out into four main approaches. A company can be "Transparent," treating openness as a core strength. It can choose to be "Secret," purposefully hiding details to protect its intellectual property. It might be "Distracting," burying the public in irrelevant data, or, it can use a "Withheld" strategy, which means either hiding bad practices or simply failing to collect the data in the first place.

When handled proactively, being transparent offers real financial and structural payoffs. For instance, Gao et al. (2023) point out that sharing detailed supply chain data sends a strong signal that a company is truly committed to long-term sustainability, which in turn improves how efficiently the company invests its capital. Beyond just signaling, the physical changes companies make to build sustainable supply chains directly boost their overall performance. Specifically, Wang et al. (2025) show that supply chain diversification drastically improves a company's Environmental, Social, and Governance (ESG) scores. Working with a wider variety of global suppliers forces large enterprises to upgrade to greener, more advanced digital tools. Interestingly, these benefits are even stronger when the market is turbulent and unpredictable (Wang et al., 2025).

In addition, She (2022) demonstrated that companies directly targeted by supply chain due diligence directives actually elevate their supply chain checks, which directly leads to better human rights practices among their immediate suppliers. However, we note that the directive itself isn't what primarily drives this success. Instead, the real changes happen when outside groups, such as NGOs and socially conscious investors, take the shared information and use it to actively hold companies accountable (She, 2022).

Simultaneously, forcing companies to disclose this information often triggers unintended market panic and superficial corporate responses. For instance, when the CTSCA was passed, Birkey et al. (2018) noticed a sharp drop in stock prices, especially for massive corporations and those in high-risk industries. Investors basically viewed the new transparency rules, and the potential for public scandals, as a major financial liability. Because of this market punishment, a lot of corporate compliance has been purely symbolic. Companies with the riskiest supply chains tend to release long, detailed, but ultimately empty reports just to protect themselves from public backlash (Birkey et al., 2018).

We see the exact same pattern playing out with the French Duty of Vigilance law. Schilling-Vacaflor and Gustafsson (2024) point out that companies take advantage of the law's broad wording to practice selective transparency. They report on easy, quantifiable environmental targets, but do not share information on potential human rights issues. Moreover, because the French law carries the threat of civil lawsuits, companies have become more guarded about what they share. Instead of fixing actual problems on the ground, they hide behind a "due diligence defense", by doing just enough paperwork to prove they are legally compliant (Schilling-Vacaflor and Gustafsson, 2024).

Overall, while existing research provides solid evidence that a company's ESG outcomes generally improve after new regulations are introduced, it largely overlooks the practical reality of how these changes actually happen. Specifically, the current literature leaves a significant gap regarding how external reporting mandates force businesses to reorganize their internal information systems. We seek to provide more insights into the softwares, data architectures, and internal processes that firms adopt to track and manage their compliance. This information is useful in further understanding the impact of supply chain due diligence directives.

5 Data Sources and Description

The primary data source for this analysis is the Computer Intelligence Database, which is provided by the commercial market research firm Aberdeen. Aberdeen conducts detailed surveys with IT managers to record the specific hardware and software deployed at their locations. A major advantage of this database is its high level of quality and timeliness. Aberdeen supplies this market data to leading technology companies, such as IBM, Cisco, and Dell, who use it to guide their business operations. Because these large firms rely heavily on this information, any errors or outdated records would be quickly discovered during their routine sales calls. This built-in commercial feedback loop ensures that the data remains accurate and highly reliable for our research.

To make sense of thousands of individual software entries, we use a three-tiered hierarchy: **Manufacturer, Model Group, and Model**. In this report, we focus on the middle tier, which describes what the software does exactly, without disclosing who made it or the names of specific products. This classification groups products based on their primary utility. In our specific data work, the Model Group is the primary lever used to identify whether a firm is meeting compliance standards, as it highlights the specific capabilities present within the organization.

6 Methodology

In order to answer the two core research questions, we must construct our data-set such that it contains all relevant information. This report aids this process by exploring and assessing the Computer Intelligence Database. We start with a list of 224 model groups and a list of 10 base business activities: Accounting/Finance, Business Administration, Customer Relationship Management, Decision Making, Document Management, Employee Collaboration, Human Resources, Information Technology, Project Management and Supply Chain Management.

Our ultimate goal is to generate a list of model groups specifically relevant to companies affected by the California Transparency in Supply Chains Act. To reach this goal, we first test whether these model groups can be clearly organized into business areas. We examine each group and identify accurate definitions through specialized dictionaries and online research. Based on these findings, we determine the most suitable business category for each candidate in the list.

Following this, we create specific subcategories. While the main categories represent standard business functions, the subcategories are designed based on our specific data to best describe the resources we have at hand. This process results in the following:

1. **Accounting/ Finance:** This category encompasses software designed to manage a company's financial transactions, payroll, banking activities, and financial forecasting. It is crucial for maintaining accurate financial records, ensuring regulatory compliance, and providing data-driven insights for economic planning. It's corresponding subcategories are:

- Core Accounting and Payroll
- Banking and Finance
- Financial Analysis and Forecasting

2. **Business Administration:** This software category provides tools for overseeing daily corporate operations, optimizing routine workflows, and managing organizational risk and governance. It is highly useful for streamlining administrative tasks and ensuring that vital information flows efficiently across different departments. It's corresponding subcategories are:

- Administrative Tasks
- Optimizing Workflows
- Information Sharing
- Risk Management and Governance

3. **Customer Relationship Management:** CRM platforms manage a company's interactions with current and potential buyers by integrating sales, marketing, and support services. They are essential for driving sales growth, improving customer retention, and maintaining a cohesive e-commerce presence. It's corresponding subcategories are:

- Sales and Marketing
- Customer Support Services
- E-Commerce, Portals and Web-Presence

4. **Decision Making:** This category refers to business intelligence and data analysis tools that process large organizational datasets to uncover trends and patterns. It is vital because it empowers leadership to make informed, evidence-based strategic choices rather than relying solely on expert judgment. It's corresponding subcategories are:

- Data Analysis
- Business Intelligence

5. **Document Management:** These systems serve as centralized, secure digital repositories for creating, storing, managing, and retrieving company documents and databases. They are important for reducing paper clutter, ensuring data security, and allowing employees to quickly find the exact records they need. It's corresponding subcategories are:

- Document and Graphic
- Database Management

6. **Employee Collaboration:** This category includes communication platforms and virtual teamwork facilities that connect staff members across different locations. It is incredibly useful for fostering teamwork, speeding up internal problem-solving, and keeping distributed teams aligned on shared goals. It's corresponding subcategories are:

- Communication Facilities
- Teamwork Facilities

7. **Human Resources:** HR software manages the entire employee life-cycle, from core benefits administration to corporate training programs and productivity tracking. It is critical for on-boarding top talent, maintaining employee satisfaction, and ensuring compliance with internal policies and labor laws. It's corresponding subcategories are:

- Core HR and Benefits
- Corporate Training
- Office Productivity Suites

8. **Information Technology:** As a software category, IT encompasses the tools used to manage, secure, deploy, and monitor an organization's internal technological infrastructure and networks. It is foundational for modern businesses to protect against cyber threats, ensure system uptime, and support the technical needs of all other departments. It's corresponding subcategories are:

- Core Systems
- Cloud and Security
- Application Development

9. **Project Management:** Project management software provides frameworks for planning initiatives, allocating personnel and resources, and tracking progress against deadlines. It is highly important for keeping complex, multi-step projects on budget and on schedule while keeping team members organized. It's corresponding subcategories are:

- General Project Management
- Resource Management

10. **Supply Chain Management:** This category includes tools for overseeing the end-to-end flow of goods, from inventory and plant manufacturing to logistics and final order processing. It is indispensable for minimizing physical delays, reducing operational costs, and ensuring that products reach customers reliably. It's corresponding subcategories are:

- Inventory Management
- Logistics and Order Processing
- Manufacturing and Plant Operations

Additionally, we identify 7 software categories that are particularly useful for companies directly affected by the CTSCA. These categories differ from the 10 general groups presented above because they have a highly specific scope. We constructed this new list based on the exact tasks a company must complete to comply with the CTSCA.

1. **Team/ Committee Communication:** This type of software provides secure digital workspaces, such as messaging boards and file-sharing hubs, allowing dedicated internal task forces to collaborate seamlessly. Following the passage of the CTSCA, many companies form these specialized committees specifically to oversee their supply chain transparency efforts. Using this software is essential for these teams to coordinate legal disclosures, react to sudden compliance issues, and manage their workflow without miscommunication.

2. **Audit:** Audit software offers specialized tools to schedule, track, and record the results of both internal and external facility inspections across global operations. A core requirement of the CTSCA is that companies must publicly disclose how extensively they audit their suppliers for anti-slavery and human trafficking compliance. Therefore, this software acts as the official system of record to prove the company is actively inspecting its supply chain and avoiding high-risk regions.

3. **Evaluation and Compliance:** This category utilizes risk frameworks and digital checklists to continuously assess supplier performance and immediately flag any violations of corporate ethical codes. Under the CTSCA, businesses are required to disclose if they demand direct suppliers to certify that their materials comply with human trafficking laws. Evaluation software manages these mandatory certifications and ensures that all partners meet necessary ethical benchmarks before business contracts are signed or renewed.

4. **Connecting Businesses in the Supply Chain:** These platforms create a digital network that links a company directly to its vendors and suppliers, establishing a clear line from raw material extraction to the final product. To make the accurate public disclosures required by the Act, a company must first know exactly who is operating within its broader supply chain. This software provides the essential visibility needed to trace product origins and verify that ethical standards are upheld even at the lowest, most obscured tiers of production.

5. **Information Sharing:** This software facilitates the secure and transparent exchange of critical corporate policies and operational data between internal departments and external supply chain partners. Transparency is the fundamental goal of the CTSCA, meaning companies must actively distribute their ethical guidelines and collect compliance data in return. This software ensures that legal expectations are clearly communicated across the supply chain and that the required disclosure data is easily accessible to the internal compliance team.

6. **HR and CRM:** This category merges internal staff management tools, which track mandatory anti-trafficking training, with external communication systems designed to log inquiries from consumers and NGOs. The CTSCA explicitly requires companies to disclose their internal accountability standards and the specific training they provide to employees. Furthermore, because the Act empowers the public to question companies about their ethical practices, the CRM component ensures the business can effectively track and respond to these external demands for transparency.

7. **Generating Reports:** Reporting software automatically pulls data from various business systems to

compile, format, and publish highly detailed compliance documents and public-facing statements. The ultimate requirement of the CTSCA is the publication of a clear, easily accessible disclosure statement on the company’s website detailing its ongoing anti-slavery efforts. This software is essential for synthesizing all the audit, training, and compliance data collected throughout the year into the exact legal formats required by the Act.

7 Results

Table 2 provides a clear breakdown of the 224 model groups, sorting them into the ten distinct categories. We see that companies prioritize foundational digital infrastructure above all else, with Information Technology and Document Management collectively accounting for exactly half of all software (50 percent). This heavy concentration indicates that before a business can focus on specialized tasks, it must invest heavily in the essential digital backbone required to secure networks, maintain hardware, and safely store corporate records. Once this foundation is established, businesses direct their investments toward the core operational tools needed to keep the company moving and profitable. This is evidenced by Supply Chain Management, Customer Relationship Management, and Accounting/Finance making up roughly a third of the dataset (approximately 32 percent). In contrast, tools focused on internal staff management, such as Employee Communication (4.91 percent) and Human Resources (3.57 percent), represent a much smaller share of the overall software count. Finally, the least common categories are highly specialized tools like Project Management (1.79 percent) and Decision Making (1.34 percent). However, these low counts do not signify a lack of importance, but rather reflect a restricted user base. While an entire workforce requires standard IT and document access, only a small fraction require access to high-level strategic software.

Table 2: Software Categories by Percentage and Count

Software Category	Percentage (%)	Count
Accounting/ Finance	9.82	22
Business Administration	6.25	14
Customer Relationship Management	10.71	24
Decision Making	1.34	3
Document Management	15.18	34
Employee Communication	4.91	11
Human Resources	3.57	8
Information Technology	34.82	78
Project Management	1.79	4
Supply Chain Management	11.61	26
Total	100.00	224

Table 3 provides a detailed list of model groups that companies affected by the CTSCA may invest in. In summary, this software list describes how organizations can remain highly structured. Under the CTSCA, they should deeply integrate their manufacturing, supply chains, and employee communication to remain transparent, compliant, and ready for strict external audits at any time.

Table 3: Relevant Model Groups and Categories

Model Group	Description	Category
REPORT-GEN	Report generation tools that extract, query, and format data from underlying databases to build customizable operational reports.	Generating Reports

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Model Group	Description	Category
REPORT-MGMT	Enterprise reporting software used to securely distribute, archive, and manage user access to organizational reports and dashboards.	Generating Reports
REPORTING	Broad data visualization and reporting platforms used to track Key Performance Indicators (KPIs) and business metrics.	Generating Reports
GROUPWARE	Groupware Software. Tracked extensively in the databases (often alongside email messaging software), this refers to collaborative software suites used to facilitate communication, scheduling, and task-sharing among employees.	Information Sharing
INTER-PROG	Middleware and integration tools that allow different software programs and enterprise systems to communicate and share data.	Information Sharing
KNOWLEDGEMGT	Knowledge Management software helps organizations capture, organize, and share internal expertise, FAQs, and documentation.	Information Sharing
SAAS-STORAGE	Cloud storage solutions (like Dropbox or Google Drive) that provide secure, remote file hosting and sharing capabilities as a service.	Information Sharing
TP-MONITOR	Transaction Processing Monitors manage and route the transfer of data and requests between client terminals and backend databases to ensure system stability.	Information Sharing
WAN-OPT-SRV	Wide Area Network (WAN) optimization software used to maximize data transfer efficiency and reduce latency across geographically dispersed networks.	Information Sharing
ENVIRONMENT	Environmental software refers to digital platforms and tools designed to monitor, manage, and report on environmental impact data, such as carbon emissions, waste, and resource consumption.	Evaluation & Compliance
GRC-SOFTWARE	Governance, Risk management, and Compliance software. Platforms used by enterprises to manage regulatory compliance, internal corporate governance, and risk mitigation strategies.	Evaluation & Compliance
SOCIAL-SVCS	Case management systems used by government and non-profit agencies to track social welfare, child welfare, and public assistance.	Evaluation & Compliance
SUSTAINABILI	Software designed to track, measure, and report on corporate sustainability metrics, carbon footprint, and ESG (Environmental, Social, Governance) compliance.	Evaluation & Compliance
EAI	Enterprise Application Integration Software (also referred to as Application Consolidation software). These systems are used to connect and integrate various disparate enterprise software applications so they can share data and business rules.	Connecting Businesses in the Supply Chain
PARTNER-MGT	Partner Relationship Management (PRM) software used to manage communication, sales channels, and interactions with external business partners.	Connecting Businesses in the Supply Chain

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Table 3 – continued from previous page

Model Group	Description	Category
SUPPLY-CHAIN	Supply Chain Management (SCM) platforms that oversee the entire flow of goods, data, and finances from raw materials to final delivery.	Connecting Businesses in the Supply Chain
SUPPLYCHAIN	Supply Chain Management (SCM) platforms that oversee the entire flow of goods, data, and finances from raw materials to final delivery.	Connecting Businesses in the Supply Chain
CONTACTCENTE	Call or Contact Center Software. Systems tracked for managing inbound and outbound customer communications, utilizing technologies like Automatic Call Distributors (ACD) and Interactive Voice Response (IVR).	HR & CRM
CRM/SFA	Customer Relationship Management / Sales Force Automation Software. A major software category used by businesses to manage customer data, track sales interactions, and automate sales pipelines.	HR & CRM
CUSTOMER-SVC	Customer Service Software. Tracked as an internet application and core business activity, these are systems and portals dedicated to providing technical support and managing customer inquiries.	HR & CRM
HUMAN-RES	Human Resource Software. Tracked individually and as the "HCM/HR Module" within broader Enterprise Resource Planning (ERP) software suites, used to manage employee data, benefits, and payroll.	HR & CRM
INSTRUCTION	Educational and instructional software used to deliver training modules, courses, and e-learning materials to employees or students.	HR & CRM
PERFORM-MEAS	Performance measurement and management tools used by HR to track employee goals, appraisals, and overall workforce productivity.	HR & CRM
PERSONNEL	Human Resource Information Systems (HRIS) used to securely store and manage core employee records, onboarding, and personnel data.	HR & CRM
SAAS	Software as a Service; a cloud computing model where generalized software applications are hosted by a vendor and accessed via the internet.	HR & CRM
SAAS-EMAIL	Cloud-hosted email and productivity suites (like Google Workspace or Microsoft 365) delivered as a subscription service.	HR & CRM
SAAS-CRM	Cloud-based Customer Relationship Management platforms (like Salesforce) that require no on-premise installation to track leads and sales.	HR & CRM
SAAS-ERP	Cloud-based Enterprise Resource Planning systems that allow organizations to manage core financial and supply chain processes online.	HR & CRM
COLLABORATIV	Asset-Mgmt & Asset Management Software. Used by enterprises to track, inventory, and manage their physical, financial, or IT assets throughout their lifecycle.	Team/ Committee Communication
COMMUNICATNS	Communications Software / Services.	Team/ Committee Communication

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Model Group	Description	Category
HANDSET-OS	Handset or Smartphone Operating System. The databases track this metric to identify the specific operating systems running on mobile handsets and smartphones provided to corporate employees.	Team/ Committee Communication
INT-OFFICE	Inter-office communication tools that facilitate seamless messaging, collaboration, and file sharing among internal employees.	Team/ Committee Communication
M-MIDDLEWARE	Mobile middleware bridges the gap between mobile applications and backend enterprise systems, ensuring seamless data synchronization.	Team/ Committee Communication
UNIFIED-COMM	Unified Communications as a Service (UCaaS) platforms that integrate voice, video, messaging, and presence into a single interface.	Team/ Committee Communication
VIDEO-CONF	Software (like Zoom or Microsoft Teams) that allows multiple users to conduct live audio-visual meetings over the internet.	Team/ Committee Communication
WORKGROUP	Collaborative groupware and project management tools designed to help small teams or departments share files and coordinate on tasks.	Team/ Committee Communication
AGRICULTURE	As a software category, this refers to Ag-Tech or farm management software used for crop tracking, livestock management, and agricultural supply chains.	Audit
ASSET-MGMT	Asset Management Software. Used by enterprises to track, inventory, and manage their physical, financial, or IT assets throughout their lifecycle.	Audit
AUDIT	Auditing software. Used by compliance and accounting teams to facilitate internal and external financial, IT, or regulatory audits.	Audit
CONSTRUCTION	Construction Management software. Used by builders and contractors for project management, material estimating, job costing, and scheduling.	Audit
DISCRETE-MFG	Discrete Manufacturing software. Specialized ERP or manufacturing execution systems used to track and manage the production of distinct, countable items (like cars, smartphones, or furniture).	Audit
DISTRIBUTION	Distribution software. Systems (often tied to Supply Chain Management) used by wholesale distributors to manage inventory, warehousing, logistics, and order fulfillment.	Audit
ERP	Enterprise Resource Planning Software. A major enterprise software category consisting of integrated modules used to manage an organization's core business processes.	Audit
ERP-UNSPEC	Enterprise Resource Planning Software. A major enterprise software category consisting of integrated modules used to manage an organization's core business processes.	Audit
FA	Fixed Assets software. A financial accounting software module used to track the lifecycle, depreciation, and value of a company's physical assets (like machinery, vehicles, and buildings).	Audit

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Model Group	Description	Category
FACILITY-MGT	Facility Management software. Computer-Aided Facility Management (CAFM) tools used to plan, maintain, and manage physical building spaces, maintenance work orders, and real estate assets.	Audit
FIXED-ASSETS	Fixed Assets software. A financial accounting software module used to track the lifecycle, depreciation, and value of a company's physical assets (like machinery, vehicles, and buildings).	Audit
G/L	General Ledger software. The core module of any accounting software system where all financial transactions, credits, and debits are recorded.	Audit
GIS	Geographic Information Systems software. Applications used to store, analyze, manipulate, and visualize geographic and spatial mapping data.	Audit
GL	General Ledger software. The core module of any accounting software system where all financial transactions, credits, and debits are recorded.	Audit
INTG-ACCT	Integrated accounting systems combine various financial modules (AP, AR, General Ledger) into one cohesive platform.	Audit
INVENTORY	Software used to track stock levels, monitor sales, and manage the lifecycle of goods from purchase to fulfillment.	Audit
INVENTORY-MG	Advanced inventory management, often integrating warehouse management systems (WMS) to optimize storage and order picking.	Audit
MANUFACTURE	Manufacturing Execution Systems (MES) that track, document, and control the transformation of raw materials into finished goods on the factory floor.	Audit
MANUFACTURER	Manufacturing Execution Systems (MES) that track, document, and control the transformation of raw materials into finished goods on the factory floor.	Audit
MAPPING	Geographic Information Systems (GIS) used to capture, store, analyze, and visualize spatial and geographic data.	Audit
MATERIAL-MGT	Material Requirements Planning (MRP) software used to manage manufacturing processes, calculate material needs, and schedule deliveries.	Audit
MRP/MANUF	Material Requirements Planning (MRP) software used to manage manufacturing processes, calculate material needs, and schedule deliveries.	Audit
ORDER-ADMIN	Back-office software used to oversee customer accounts, manage pricing tiers, and configure order rules before processing.	Audit
ORDER-PROC	Systems that capture, track, and fulfill customer orders in real-time, bridging sales channels with inventory and shipping.	Audit
PIM	Product Information Management (PIM) software acts as a central repository to manage and enrich product data for e-commerce and catalogs.	Audit
PLANT-MGMT	Software systems used to oversee overall facility health, maintenance schedules, and resource allocation across a manufacturing plant.	Audit

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Table 3 – continued from previous page

Model Group	Description	Category
PLANT-OPS	Operational control systems, like SCADA (Supervisory Control and Data Acquisition), used for real-time monitoring and control of plant machinery.	Audit
PROCESS-MFG	Specialized ERP and manufacturing software designed for recipe- or formula-based production (e.g., food, beverages, chemicals).	Audit
PROCUREMENT	Procure-to-pay software that automates the purchasing of goods and services, vendor selection, and purchase order tracking.	Audit
PROJECT-MGMT	Task and resource management software used to plan schedules, assign duties, and track the day-to-day progress of individual projects.	Audit
PROJECT-MGT	Task and resource management software used to plan schedules, assign duties, and track the day-to-day progress of individual projects.	Audit
RESOURCE-MGT	Enterprise resource management software used to forecast, allocate, and schedule personnel, equipment, and budgets for projects.	Audit

8 Conclusion

Overall, the primary objective of this report was to conduct a qualitative review of existing literature and proprietary data.

The initial section focused on understanding the California Transparency in Supply Chains Act of 2012, examining both its legislative architecture and the subsequent corporate reactions. To provide a comprehensive context, we compared this legislation to European frameworks, including France’s Loi de Vigilance, Germany’s Lieferkettensorgfaltspflichtengesetz, and the UK’s Modern Slavery Act. This comparative analysis highlighted a definitive evolution in regulatory approaches. Older mandates, such as the CTSCA and the UK’s Modern Slavery Act, function primarily on passive disclosure. They encourage ethical operations by relying on consumer choices, investor pressure, and the threat of public shaming rather than imposing direct legal penalties for poor practices. Conversely, recent European directives adopt a different framework. They legally mandate large enterprises to actively identify, mitigate, and prevent human rights and environmental abuses deep within their multi-tier supply chains. To enforce compliance, these modern frameworks utilize severe mechanisms, including substantial financial penalties tied to global revenue, civil and criminal liability, and exclusion from public procurement contracts.

The subsequent sections of the report explored possible software categories that companies can employ to meet these rigorous regulatory demands. We identified seven general software categories that appear particularly useful for companies directly affected by the CTSCA: Team/Committee Communication, Audit, Evaluation and Compliance, Connecting Businesses in the Supply Chain, Information Sharing, HR and CRM, and Generating Reports. Furthermore, we analyzed our data on Model Groups and created a detailed list of 68 relevant items.

This classification illustrates the specific digital tools companies can utilize to maintain structured and interconnected operations. To effectively comply with mandates like the CTSCA, corporations should deeply integrate their manufacturing processes, supply chain networks, and internal employee communications. This level of digital synchronization is useful for firms to ensure transparency and remain consistently prepared for external audits at any time.

Ultimately, before any broader research question can be effectively answered, a crucial first step is simply determining whether the necessary data is available. By actively exploring the dataset, categorizing the various software systems, and carefully identifying which specific tools are relevant for managing regulatory

compliance, we try to provide a clear map of the available resources. In order to adequately support further research, this paper provides preliminary results, describes the 2012 CTSCA and outlines potential software integrations, concluding that the Computer Intelligence Database is a feasible data source for the core research questions.

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